

Safety Data Sheet

Issue Date: April 6, 2015

Revision Date: May 2, 2016

Version 2

1. IDENTIFICATION

Product Identifier

Product Name Steric P+K DS

Other means of identification

SDS # FFN: 00907

UN/ID No UN1479

Recommended use of the chemical and restrictions on use

Recommended Use Manufacture Use Concentrate.

Details of the supplier of the safety data sheet

Supplier Address

Verdesian Life Sciences, U.S., LLC.
1001 Winstead Drive, Suite 480
Cary, NC 27513

Emergency Telephone Number

Company Phone Number Business Phone: (800) 868-6446
Fax Phone: (919) 535-3652
Emergency Telephone (24 hr) INFOTRAC 1-352-323-3500 (International)
1-800-535-5053 (North America)

2. HAZARDS IDENTIFICATION

Appearance Light green crystals

Physical State Solid

Odor None

Classification

Oxidizing Solids	Category 3
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Hazards Not Otherwise Classified (HNOC)

May be harmful if swallowed

Signal Word

Warning

Hazard Statements

May intensify fire; oxidizer



Precautionary Statements - Prevention

Keep away from heat/sparks/open flames/hot surfaces. — No smoking
Keep/Store away from clothing/Incompatible/combustible materials
Take any precaution to avoid mixing with combustibles/Incompatible
Wear protective gloves/protective clothing/eye protection/face protection

Precautionary Statements - Response

IN CASE OF FIRE: Use CO2, dry chemical, or foam for extinction

Precautionary Statements - Disposal

Dispose of contents/container to an approved waste disposal plant

3. COMPOSITION/INFORMATION ON INGREDIENTS

Chemical Name	CAS No	Weight-%
Mono-ammonium Phosphate	7722-76-1	Proprietary
Potassium Nitrate	7757-79-1	Proprietary
Potassium Sulfate	7778-80-5	Proprietary
Potassium Chloride	7447-40-7	Proprietary
Ammonium Sulfate	7783-20-2	Proprietary

If Chemical Name/CAS No is "proprietary" and/or Weight-% is listed as a range, the specific chemical identity and/or percentage of composition has been withheld as a trade secret.

4. FIRST-AID MEASURES**First Aid Measures**

General Advice	Provide this SDS to medical personnel for treatment.
Eye Contact	If contact with eyes, immediately flush eyes with plenty of water for at least 15 minutes lifting upper and lower eyelids occasionally. If eye irritation persists: Get medical advice/attention.
Skin Contact	Wash with soap and water. Take off contaminated clothing. Wash contaminated clothing before reuse. Get medical attention if irritation develops or persists.
Inhalation	Remove exposed individual(s) to fresh air for 20 minutes. Consult a physician / poison center if individual's condition declines or if symptoms persist.
Ingestion	Do not induce vomiting without medical advice. Rinse mouth. Never give anything by mouth to an unconscious person. Call a physician or poison control center immediately.

Most important symptoms and effects

Symptoms	May cause eye, skin, and respiratory tract irritation.
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Indication of any immediate medical attention and special treatment needed

Notes to Physician	Treat symptomatically and supportively. Persons with pre-existing skin disorders or impaired respiratory function may be more susceptible to the effects of the substance.
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5. FIRE-FIGHTING MEASURES

Suitable Extinguishing Media

Dry chemical. Water spray (fog). Carbon dioxide (CO₂).

Unsuitable Extinguishing Media Not determined.

Specific Hazards Arising from the Chemical

May intensify fire; oxidizer. Emits toxic fumes under fire conditions.

Hazardous Combustion Products Oxides of nitrogen.

Protective equipment and precautions for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

6. ACCIDENTAL RELEASE MEASURES

Personal precautions, protective equipment and emergency procedures

Personal Precautions Ventilate affected area. Use personal protection recommended in Section 8.

Environmental Precautions Do not flush into surface water or sanitary sewer system. See Section 12 for additional Ecological Information.

Methods and material for containment and cleaning up

Methods for Containment Prevent further leakage or spillage if safe to do so.

Methods for Clean-Up Avoid creating dust. Sweep up and shovel into suitable containers for disposal. Discard any product, residue, disposable container or liner in full compliance with federal, state, and local regulations.

7. HANDLING AND STORAGE

Precautions for safe handling

Advice on Safe Handling Handle in accordance with good industrial hygiene and safety practice. Use personal protection recommended in Section 8. Use with adequate ventilation. Avoid contact with skin, eyes or clothing. Do not eat, drink or smoke when using this product. Wash face, hands, and any exposed skin thoroughly after handling. Avoid generation of dust. Keep away from heat/sparks/open flames/hot surfaces. — No smoking. Keep/Store away from clothing/Incompatible/combustible materials. Take any precaution to avoid mixing with combustibles.

Conditions for safe storage, including any incompatibilities

Storage Conditions Keep containers tightly closed in a dry, cool and well-ventilated place.

Incompatible Materials Strong alkalis. Strong acids. Reducing agent. Combustible material.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Exposure Guidelines No exposure limits noted for ingredient(s). The following information is given as general guidance

Appropriate engineering controls

Engineering Controls Maintain eye wash fountain and quick-drench facilities in work area.

Individual protection measures, such as personal protective equipment

Eye/Face Protection Use safety goggles, safety glasses with side shields and/or a full face shield as described by OSHA's eye and face protection regulations in 29CFR 1910.133.

Skin and Body Protection Wear impervious protective clothing, including boots, gloves, lab coat, apron or coveralls, as appropriate, to prevent skin contact.

Respiratory Protection Follow OSHA respirator regulations (29 CFR 1910.134) and, if necessary, wear a MSHA/NIOSH-approved respirator.

General Hygiene Considerations Avoid contact with skin, eyes and clothing. After handling this product, wash hands before eating, drinking, or smoking. If contact occurs, remove contaminated clothing. If needed, take first aid action shown on section 4 of this SDS. Launder contaminated clothing before reuse.

9. PHYSICAL AND CHEMICAL PROPERTIES

Information on basic physical and chemical properties

Physical State	Solid	Odor	None
Appearance	Light green crystals	Odor Threshold	Not determined
Color	Light green		

<u>Property</u>	<u>Values</u>	<u>Remarks • Method</u>
pH	Not applicable	
Melting Point/Freezing Point	Not determined	
Boiling Point/Boiling Range	Not determined	
Flash Point	Not determined	
Evaporation Rate	Not determined	
Flammability (Solid, Gas)	Not determined	
Upper Flammability Limits	Not determined	
Lower Flammability Limit	Not determined	
Vapor Pressure	Not determined	
Vapor Density	Not determined	
Specific Gravity	59.7 lb/f3	
Water Solubility	Not determined	
Solubility in other solvents	Not determined	
Partition Coefficient	Not determined	
Auto-ignition Temperature	Not determined	
Decomposition Temperature	Not determined	
Kinematic Viscosity	Not determined	
Dynamic Viscosity	Not determined	
Explosive Properties	Not determined	
Oxidizing Properties	Not determined	

10. STABILITY AND REACTIVITY

Reactivity

Not reactive under normal conditions.

Chemical Stability

Stable under recommended storage conditions.

Possibility of Hazardous Reactions

None under normal processing.

Hazardous Polymerization

Hazardous polymerization does not occur.

Conditions to Avoid

Incompatible materials, ignition sources and excessive heat.

Incompatible Materials

Strong alkalis. Strong acids. Reducing agent. Combustible material.

Hazardous Decomposition Products

Carbon oxides.

11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure

Product Information

To the best of our knowledge, the chemical, physical and toxicological properties of this material have not been thoroughly investigated

Eye Contact

Irritating to eyes.

Skin Contact

Irritating to skin.

Inhalation

Do not inhale.

Ingestion

Ingestion may cause irritation to mucous membranes.

Component Information

Chemical Name	Oral LD50	Dermal LD50	Inhalation LC50
Mono-ammonium Phosphate 7722-76-1	= 5750 mg/kg (Rat)	> 7940 mg/kg (Rabbit)	-
Potassium Nitrate 7757-79-1	= 3015 mg/kg (Rat)	-	-
Potassium Sulfate 7778-80-5	= 6600 mg/kg (Rat)	-	-
Potassium Chloride 7447-40-7	= 2600 mg/kg (Rat)	-	-
Ammonium Sulfate 7783-20-2	= 2000 mg/kg (Rat)	-	-

Information on physical, chemical and toxicological effects

Symptoms

Please see section 4 of this SDS for symptoms.

Delayed and immediate effects as well as chronic effects from short and long-term exposure

Carcinogenicity

Nitrate or nitrite ingested under conditions that result in endogenous nitrosation are considered IARC group 2A carcinogens.

Chemical Name	ACGIH	IARC	NTP	OSHA
Potassium Nitrate 7757-79-1		Group 2A		X

Legend*IARC (International Agency for Research on Cancer)**Group 2A - Probably Carcinogenic to Humans**OSHA (Occupational Safety and Health Administration of the US Department of Labor)**X - Present***Numerical measures of toxicity**

Not determined

12. ECOLOGICAL INFORMATION**Ecotoxicity**

An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.

Component Information

Chemical Name	Algae/aquatic plants	Fish	Toxicity to microorganisms	Crustacea
Potassium Sulfate 7778-80-5	2900: 72 h <i>Desmodesmus subspicatus</i> mg/L EC50	653: 96 h <i>Lepomis macrochirus</i> mg/L LC50 3550: 96 h <i>Lepomis macrochirus</i> mg/L LC50 static 510 - 880: 96 h <i>Pimephales promelas</i> mg/L LC50 static		890: 48 h <i>Daphnia magna</i> mg/L EC50
Potassium Chloride 7447-40-7	2500: 72 h <i>Desmodesmus subspicatus</i> mg/L EC50	1060: 96 h <i>Lepomis macrochirus</i> mg/L LC50 static 750 - 1020: 96 h <i>Pimephales promelas</i> mg/L LC50 static		825: 48 h <i>Daphnia magna</i> mg/L EC50 83: 48 h <i>Daphnia magna</i> mg/L EC50 Static
Ammonium Sulfate 7783-20-2		250: 96 h <i>Brachydanio rerio</i> mg/L LC50 480: 96 h <i>Brachydanio rerio</i> mg/L LC50 flow-through 126: 96 h <i>Poecilia reticulata</i> mg/L LC50 460 - 1000: 96 h <i>Leuciscus idus</i> mg/L LC50 static 420: 96 h <i>Brachydanio rerio</i> mg/L LC50 semi-static 18: 96 h <i>Cyprinus carpio</i> mg/L LC50 32.2 - 41.9: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 flow-through 5.2 - 8.2: 96 h <i>Oncorhynchus mykiss</i> mg/L LC50 static 100: 96 h <i>Pimephales promelas</i> mg/L LC50 123 - 128: 96 h <i>Poecilia reticulata</i> mg/L LC50 semi-static		14: 48 h <i>Daphnia magna</i> mg/L LC50 423: 24 h <i>Daphnia magna</i> mg/L EC50

Persistence/Degradability

Not determined.

Bioaccumulation

Not determined.

Mobility

Chemical Name	Partition Coefficient
Ammonium Sulfate 7783-20-2	-5.1

Other Adverse Effects

Not determined

13. DISPOSAL CONSIDERATIONS**Waste Treatment Methods**

Disposal of Wastes	Disposal should be in accordance with applicable regional, national and local laws and regulations.
Contaminated Packaging	Disposal should be in accordance with applicable regional, national and local laws and regulations.

California Hazardous Waste Status

Chemical Name	California Hazardous Waste Status
Potassium Nitrate 7757-79-1	Ignitable Reactive

14. TRANSPORT INFORMATION

Note Please see current shipping paper for most up to date shipping information, including exemptions and special circumstances.

DOT

UN/ID No	UN1479
Proper Shipping Name	Oxidizing solid, n.o.s. (contains Potassium nitrate)
Hazard Class	5.1
Packing Group	III

IATA

UN/ID No	UN1479
Proper Shipping Name	Oxidizing solid, n.o.s. (contains Potassium nitrate)
Hazard Class	5.1
Packing Group	III

IMDG

UN/ID No	UN1479
Proper Shipping Name	Oxidizing solid, n.o.s. (contains Potassium nitrate)
Hazard Class	5.1
Packing Group	III

15. REGULATORY INFORMATION**International Inventories**

Chemical Name	TSCA	DSL	NDSL	EINECS	ELINCS	ENCS	IECSC	KECL	PICCS	AICS
Mono-ammonium Phosphate	Present	X		Present		Present	X	Present	X	X
Potassium Nitrate	Present	X		Present		Present	X	Present	X	X
Potassium Sulfate	Present	X		Present		Present	X	Present	X	X
Potassium Chloride	Present	X		Present		Present	X	Present	X	X
Ammonium Sulfate	Present	X		Present		Present	X	Present	X	X

Legend:

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

EINECS/ELINCS - European Inventory of Existing Chemical Substances/European List of Notified Chemical Substances

ENCS - Japan Existing and New Chemical Substances

IECSC - China Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

AICS - Australian Inventory of Chemical Substances

US Federal Regulations

CERCLA

This material, as supplied, does not contain any substances regulated as hazardous substances under the Comprehensive Environmental Response Compensation and Liability Act (CERCLA) (40 CFR 302) or the Superfund Amendments and Reauthorization Act (SARA) (40 CFR 355).

SARA 313

Section 313 of Title III of the Superfund Amendments and Reauthorization Act of 1986 (SARA). This product contains a chemical or chemicals which are subject to the reporting requirements of the Act and Title 40 of the Code of Federal Regulations, Part 372

Chemical Name	CAS No	Weight-%	SARA 313 - Threshold Values %
Mono-ammonium Phosphate - 7722-76-1	7722-76-1	Proprietary	1.0
Potassium Nitrate - 7757-79-1	7757-79-1	Proprietary	1.0
Ammonium Sulfate - 7783-20-2	7783-20-2	Proprietary	1.0

CWA (Clean Water Act)

This product does not contain any substances regulated as pollutants pursuant to the Clean Water Act (40 CFR 122.21 and 40 CFR 122.42)

US State Regulations

California Proposition 65

This product does not contain any Proposition 65 chemicals.

U.S. State Right-to-Know Regulations

Chemical Name	New Jersey	Massachusetts	Pennsylvania
Potassium Nitrate 7757-79-1	X	X	X
Ammonium Sulfate 7783-20-2		X	X

16. OTHER INFORMATION

<u>NFPA</u>	Health Hazards	Flammability	Instability	Special Hazards
	1	0	1	OX
<u>HMIS</u>	Health Hazards	Flammability	Physical Hazards	Personal Protection
	1	0	1	C

Issue Date: April 6, 2015
Revision Date: May 2, 2016
Revision Note: Address Change

Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.

End of Safety Data Sheet